



Beyond Copy-Paste: Using AI to Think Better, Not Think Less

Dr Mastaneh Davis

mastaneh.davis@roehampton.ac.uk

Office: DB427



What we will explore together

- How do you use AI: The Good, the bad, the copy-paste
- Prompt engineering : Getting AI to work for your learning
- Becoming an AI detective: spotting garbage
- Critical Thinking in the age of AI
- Practical tools

Ground Rules

- We ALL use AI (that's okay 😊)
- Share your experience honestly
- Learn from each other

Activity: Can you please complete the following questions?

[Student Perspectives on AI Tools, AI Skills and Preparation for AI in the Workplace! – Fill in form](https://forms.office.com/e/EK5qFV1Sm7)

<https://forms.office.com/e/EK5qFV1Sm7>

**Student Perspectives on AI Tools,
AI Skills and Preparation for AI in
the Workplace!**



Can you spot the fake?

- **Game: Which picture is real? Which is AI generated?**
 - This game is a project by the University of Washington that challenges you to spot which image is real and which one is computer-generated.
- 🎮 **Game Rules:**
 - Stand up if you think there is a FAKE Image and select which one.
 - Sit down if you think there is NO FAKE Image
- <https://artsandculture.google.com/experiment/odd-one-out/wAHNn4JsVTFOiw?hl=en>



The Reality Check

- Team Challenge:
 - In your teams, think of as many answers as possible!
 - Then share one answer at a time with the class
- **Questions:**
- **What CAN AI DO?**
- **What CAN'T AI Do?**

The Reality Check

What AI Can do!

- Explain concepts
- Brainstorming
- Check your work
- Answer questions
- Save time
- Help with programming and coding

What AI Can't do!

- Think for you
- Guarantee accuracy
- Understand YOUR unique perspective
- Cite resources reliably
- Replace deep learning

What is prompt and What is Prompting?

provide a short description for what is prompt and what is prompting

 Copilot ⓘ

Here's a concise explanation:

- **Prompt:** A prompt is the input or instruction given to an AI model (or any system) to generate a response. It can be a question, a statement, or a set of guidelines that directs the model on what to produce.
- **Prompting:** Prompting is the process of crafting and providing those instructions effectively to guide the AI toward the desired output. It involves choosing the right words, context, and structure to influence the model's response.

Weaknesses

DALL-E 5.2 ▾

🎁 Free offer ×

draw a nurse



Made with the old version of image generation. [Learn more](#)

I've drawn a nurse for you 😊

Free ChatGPT

Weaknesses

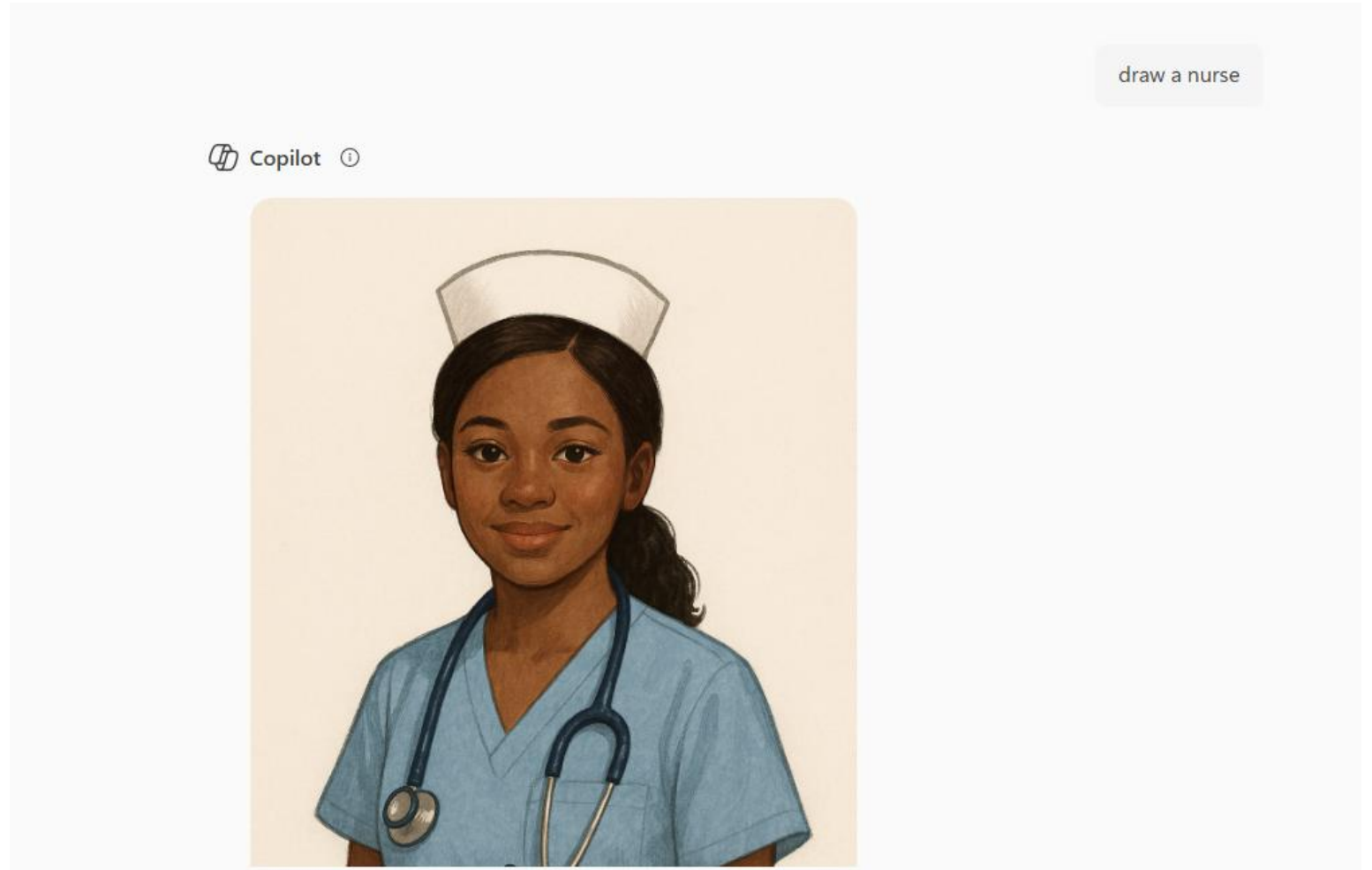
draw a nurse



Ask Gemini 3

Gemini AI

Weaknesses



Copilot

Basic Prompt Strategies

Role

Context

Task

Format

Rules

Examples

Activity – Identify strengths and weaknesses in each prompt

- **Prompt 1** - write about advancement in digital forensics
- **Prompt2** - write about advancement in digital forensics in 300 words including reliable references
- **Prompt 3** - Think critically and as a researcher then write about advancement in digital forensics focusing on identifying research gap in mobile application. This should be around 300 words including verifiable, peer-reviewed sources with full citations.

Can we improve it further?

- Act as an **expert** write about recent advancements in **mobile forensics in 500 words**, specifically:
 - Identify current research gaps in mobile application forensics
 - Analyse the challenge of inconsistent policies for mobile application forensics
 - Critically evaluate why standardisation is lacking
- Include **peer-reviewed** sources from **2020-2024** with FULL citations (author, year, journal, title) in **Harvard format** so I can verify them.
- After the review, list **3 questions** I should research further to deepen my understanding. Be **concise and critical. Avoid generic statements.**

Improve the AI Output!

Improved Prompt Example:

**Specification audience
(Students, Staff, ..)**

**Lists exact
requirements**

Asks for examples

Clear format needed



Key Learning: Don't accept the first AI output! Always check quality and refine your prompts if needed. YOU are in control of the quality!

Prompt Evolution

- Prompt example:
 - Write about climate change
 - Explain the main causes of climate change
 - Explain three main human causes of climate change, with specific examples
 - Act as an environmental science tutor. Explain three main human causes of climate change with specific examples. **After each cause, ask me a question to check my understanding before moving to the next one.** Keep explanations clear for an undergraduate level.

Phrases to Test

- Think step-by-step about this
- Critically analyse both positions
- Provide specific evidence for each claim
- What are the limitations or weaknesses in each argument?
- Cite real sources that I can verify
- Challenge common assumptions about this topic

Fact Checking - Red Flag Detection

- Vague claims without specifics ("studies show...", "experts say...")
- Citations without full details (author, year, journal)
- Statistics without sources, perfect round numbers (exactly 50%)
- Claims that seem too convenient or perfect
- Inconsistencies within the same response
- Overly confident language with no caveats

Prompt Engineering Strategies

Moving Beyond Copy-Past

- AI as a **thinking partner** rather than an **answer machine**
 - The real learning happens in the dialogue, not the output
 - AI responses are starting points for their own thinking, not endpoints
 - Your unique insights, analysis, and synthesis are what create value

Key Takeaways: Copyright & Ethics:

AI is powerful but has limitations

Clear, specific prompts get better results

AI can create many types of content

Always think critically about AI outputs

Be aware of copyright and ethical use

Key Takeaway

- AI-generated content may have copyright issues
- Always check if you can use AI outputs for your purpose
- Credit sources and be transparent about AI use

Using AI Ethically

Ethical Use		Academic Misconduct
AI as a Tutor		Direct submission – Copy-pasting AI output as your own work
Brainstorming partner		Using AI's fake references without checking
Checking your work: Ask AI to critique what YOU wrote		Having AI complete entire assignments
Learning Tool – Step by step explanation		Not disclosing AI use when required

QUESTIONS?



Thank you!